Last updated in August 2026

Thisara Nethsara

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Professional Summary

A 3rd-year Mechanical Engineering undergraduate at the University of Moratuwa with a strong interest in machine design, automated manufacturing, control systems, and energy systems. Experienced in 3D CAD (SolidWorks, AutoCAD), FEA simulation (ANSYS), and dynamic modeling (MATLAB). Developed a range of projects integrating mechanical structures with electronic controls using OOP (C++, Python) and microcontrollers (Arduino, PIC). A motivated and adaptable student seeking an internship with a strong willingness to learn, solve complex engineering challenges, and apply multidisciplinary knowledge to real-world industry applications.

Education

University of Moratuwa: B.Sc Engineering (Hons) in Mechanical Engineering (UG) Aug 2024 – Present
Cumulative GPA: 3.84 / 4.00 upto semester 4

- **Dean's List** in semester 2, 3 and 4
- **Core Modules Undertaken:** Machine Design, Control Systems Engineering, Instrumentation and Automation, Manufacturing Engineering, Thermodynamics

Royal College, Panadura Jan 2014 – Feb 2023

- **G.C.E. Advanced Level:** Physical Science Stream
- **Subjects:** Combined Mathematics, Physics, Chemistry (Z-Score: 2.5094)
- **G.C.E. Ordinary Level:** 9 A's

Technical Skills

CAD & Engineering Design: SolidEdge, Solidworks, AutoCAD
Programming: Phython ,C++, OOP, Ladder programming for PLC
Software: MATLAB, Ansys, FluidSim, STEP 7 MicroWIN, tecnomatix, Microsoft Office, Excel
Manufacturing & Fabrication: Machining, Drilling, Welding, Cutting, Fabrication

Projects

Design a gearbox for Multi-Functional Rubber Crepe Rolling Machine July 2026 - Present
Team Project — 03 Members

- *Designed a multi-ratio gearbox and combined smooth/grooved roller system tailored for Sri Lankan rubber smallholders.*
- *Calculated rolling forces for non-Newtonian polymer flows using Gaskell calendering theory and the Ostwald-de Waele power law model.*
- *Conducted rigorous stress and specification calculations for power transmission elements, including pulley systems, shaft keyways, housing seals, and fluid lubrication mechanisms.*

Solid Tyre Production Layout and Material Handling Optimization in Ferentino tyre Corporation PVT LTD
July 2026 - Present
Team Project — 04 Members

- *Simulated solid tyre production workflows using Tecnomatix to analyze floor layout utilization, identify machinery bottlenecks, and minimize manual foot traffic.*
- *Evaluated the fuel, time, and labor costs of existing forklift operations to propose and cost-justify a gantry system for handling heavy Press-On-Band (POB) metal bands and tyres.*

Reverse Engineering of a Domestic Gas Burner Jan 2026 - May 2026
Team Project — 05 Members

- *Reverse-engineered gas stove components—including the burner, manifold, and electrode—by taking measurements to create accurate engineering drawings.*
- *Identified material compositions using Optical Emission Spectrometry (OES), density analysis, and Shore D hardness testing.*
- *Evaluated mass-production feasibility for sand casting, sheet metal forming, and injection molding, alongside designing appropriate core boxes and blanking dies.*

Automated Medicine Distributor with Patient Assistance System Jan 2025 - Jun 2025
Team Project — 10 Members

- Designed the mechanical chassis and structural framework for an autonomous hospital delivery robot, integrating a 4-wheel differential drive system for stable navigation.
- Designed the electronic system enabling real-time line detection and motor control for autonomous navigation along hospital corridor paths.
- Implemented PID control algorithms to achieve stable, accurate path tracking, tuning proportional, integral, and derivative gains for smooth navigation and turning at junctions.

Fabrication of a Machine Vice Jan 2025 - May 2025
Team Project — 04 Members

- Designed and fabricated a machine vice from steel rod, box bar, and plate stock, covering the full workflow from technical drawing to finished assembly.
- Performed cutting, welding, drilling, and machining operations to construct the vice body, jaws, and clamping mechanism to design tolerances

Volunteering Experience

IEEE Student Branch of University of Moratuwa Sep 2025
Technical Committee Member and Speaker — GAMMADDATA IEEE API 4.0 – 2025
Served on the Technical Committee for GAMMADDATA IEEE API 4.0, a national-level program hosted by University of Moratuwa's IEEE student chapters. Delivered a hands-on Arduino seminar for participating students, contributing to the event's goal of inspiring students across Sri Lanka in technical and applied engineering skills.

Memberships

Institute of Mechanical Engineers: *Student Affiliate Member (Membership Number: 80789155)*

References

Dr. Lihil U. Subasinghe
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Department of Mechanical Engineering
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